

Problem 1

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}$$

Solution. $\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$

Answer:

1

Problem 2

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix} \quad \det(A)$$

Solution. A 3×3

/ det1

Answer:

1

Problem 3

$$\begin{aligned} x + y + z &= 6 & 2x - y + z &= 3 & x + 2y - z &= 2 \end{aligned}$$

Solution. 3 linsolve

$\{(1, 2, 3)\}$

Answer:

$\{(1, 2, 3)\}$

Problem 4

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

Solution. $|I - A| = 0$

$$\begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix} = 31 \begin{vmatrix} 1 \\ 1 \end{vmatrix}$$

Answer:

$$\begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix} = 31 \begin{vmatrix} 1 \\ 1 \end{vmatrix}$$

Problem 5

$$y'' - 5y' + 6y = 0$$

Solution. $2 \cdot 6 \cdot y(x) - 5 \frac{d}{dx} y(x) + \frac{d^2}{dx^2} y(x) = 0$

dsolve

$$y(x) = (C_1 + C_2 e^x) e^{2x}$$

Answer:

$$y(x) = (C_1 + C_2 e^x) e^{2x}$$

Problem 6

$$g = 9.8 \text{ m/s}^2$$

Solution. $v = g t = \frac{1}{2} g t^2 = 9.8 \text{ m/s}^2$

$$t = 3.0 \text{ s} \quad v = 9.8 \times 3.0 = 29.400000000000002 \text{ m/s}$$

$$s = \frac{1}{2} \times 9.8 \times 3.0^2 = 44.1 \text{ m}$$

Answer:

$$v = 29.400000000000002 \text{ m/s} = 44.1 \text{ m}$$

Auto-generated: 6/6 problems solved by SymPy.